

STEADYHAND

Product Requirements Document

v2.0 · June 2026

One angle. One shot. Steady wins.

Document Status	Draft — v2.0
Platform	Progressive Web App (iOS + Android)
Author	Ryan
Last Updated	June 2026
Target Launch	Q3 2026
Previous Version	v1.0 — iOS Native (March 2026)

Table of Contents

1. Executive Summary

2. What's Changed from v1.0

2.1 Platform

2.2 Game Loop Improvements

2.3 Leaderboard Improvements

3. Problem & Opportunity

3.1 The Problem

3.2 The Opportunity

4. Target Audience

4.1 Primary

4.2 Secondary

5. Core Game Loop

5.1 Step-by-Step Flow

5.2 Scoring Formula

5.3 Practice Mode

6. Anti-Cheat System

7. Daily Angle System

7.1 Generation Rules

7.2 Reveal Schedule

8. Leaderboard Architecture

8.1 Three-Tier Structure

8.2 Rank Display

8.3 Friend Leaderboards

8.4 End-of-Day Medal Ceremony

9. Profile & Progression

9.1 Profile Elements

9.2 Rank Tiers

10. UX & Interface Design

10.1 Screen Inventory

10.2 Audio Design

11. Technical Architecture

11.1 Stack Overview

11.2 Sensor Specification

11.3 Backend Schema

12. Notifications

13. Monetisation Strategy

13.1 Phase 1 — Free

13.2 Phase 2 — Cosmetic IAP

13.3 Phase 3 — SteadyHand Pro

14. Success Metrics

15. Development Milestones

16. Risks & Mitigations

17. Out of Scope (v2.0)

18. Open Questions

18.1 Hold Duration

18.2 Tolerance Window

18.3 Sensor Normalisation

18.4 Android App (Future)

1. Executive Summary

SteadyHand is a daily cross-platform skill game built around a single mechanic: tilt your phone to a secret angle and hold it perfectly still. Every day, a new target angle is revealed at the moment you tap Start. You get one attempt. Your score is your average deviation in degrees during the hold window. Lower is better.

Version 2.0 transitions SteadyHand from a native iOS application to a Progressive Web App (PWA), making it available to both iOS and Android users from a single codebase without an App Store dependency. All core mechanics, anti-cheat systems, and competitive structure from v1.0 are preserved, with targeted improvements to the game loop, leaderboard, and onboarding.

The core loop is 30 seconds long. The competitive hook lasts all day.

2. What's Changed from v1.0

2.1 Platform

Area	v1.0 (iOS Native)	v2.0 (PWA)
Platform	iOS 16+ only	iOS Safari + Android Chrome
Distribution	App Store	Direct URL / Add to Home Screen
Sensor API	CoreMotion (100Hz)	DeviceOrientation API (~60Hz)
Haptics	Rich haptic API	Vibration API (Android) / Audio (iOS)
Push notifications	APNs	Web Push API (requires PWA install)
Auth	Sign in with Apple	Sign in with Apple + Google OAuth
Hosting	Xcode / TestFlight	Vercel

2.2 Game Loop Improvements

Change	v1.0 Behaviour	v2.0 Behaviour	Reason
Drift penalty	Timer pauses and resets to 10s on drift	Timer keeps running; drift weighted in MAD score	Reset felt unfair; browser sensors can spike
Angle reveal	1 second flash	1.5–2 seconds	Accounts for browser render variability
Practice mode	Not included (open question)	Included in v1.0 launch	Reduces new user churn

2.3 Leaderboard Improvements

Change	v1.0 Behaviour	v2.0 Behaviour	Reason
Rank display	Absolute position (#47 of 203)	Percentile rank (top 8%)	Motivating at any player count; solves cold-start
Weekly metric	7-day medal count	Lowest average MAD across 7 attempts	Rewards consistency, not just daily wins
Friend groups	Phase 3 — Pro subscription only	Up to 5 people free; 20 people Pro	Key daily retention driver
Tiebreaking	Earliest submission time, then account age	Tied scores share rank equally	Removes midnight-play incentive

3. Problem & Opportunity

3.1 The Problem

Mobile gaming is dominated by either long-session games (RPGs, strategy) or mindless tap-loops. There is a gap in the market for games that are:

- Extremely short to play — under 60 seconds per session
- Physically skilled — not luck-based or reflex-based
- Socially competitive in a low-pressure daily format
- Novel enough to be a conversation starter
- Accessible on any smartphone without an App Store download

3.2 The Opportunity

The gyroscope and accelerometer in every modern smartphone are almost entirely unused by consumer apps in creative ways. SteadyHand exploits a sensor that has been sitting idle — turning precision motion control into a competitive sport. The Wordle model proved that a single-attempt daily game with a global leaderboard generates organic sharing and habit formation without monetisation pressure. Moving to PWA removes the iOS-only constraint and doubles the addressable audience.

4. Target Audience

4.1 Primary

- Ages 18–35, competitive casual gamers
- Daily Wordle / NYT Games players
- People who share game scores on social media
- iPhone and Android users

4.2 Secondary

- Surgeons, artists, athletes — communities with a professional interest in hand steadiness
- Friend groups looking for a shared daily challenge
- Streamers and content creators — watchable, reactable format

5. Core Game Loop

The entire game is a single linear flow. There are no menus to navigate mid-attempt, no retries, and no distractions once the player taps Start.

5.1 Step-by-Step Flow

Step	State	Description
1	Home	Player sees today's date, personal best, and global leaderboard percentile. Target angle shown as ??°.
2	Pre-Start Gate	Player taps START. App requires 2–3 seconds of continuous phone movement to confirm device is in-hand. A visual motion ring fills as they move.
3	Angle Reveal	Target angle flashes on screen (e.g. 47°) for 1.5–2 seconds before the live tilt UI appears. Time adjusted from v1.0 for browser render variability.
4	Targeting	A live dial shows current tilt. Player tilts toward target. A tolerance arc ($\pm 2^\circ$) is shown. Timer has not started yet.
5	Lock & Hold	Once within $\pm 2^\circ$ for 0.5 seconds, hold timer starts (10 seconds). Audio cue confirms lock. Timer counts down.
6	Drift (Updated)	If player drifts outside $\pm 2^\circ$, timer keeps running. Drift moments are weighted more heavily in the MAD score. No timer reset.
7	Completion	Timer reaches zero. Score calculated as weighted MAD during the 10-second hold window.
8	Score Screen	Score displayed (e.g. 0.34°). Percentile rank shown. Share card generated. Leaderboard updates in real time.

5.2 Scoring Formula

Score = weighted mean absolute deviation (MAD) of gyroscope pitch/roll readings sampled at 50Hz during the 10-second hold window, expressed in degrees to 2 decimal places. Drift periods outside $\pm 2^\circ$ are assigned a higher weight multiplier in the MAD calculation.

Lower score = better. A perfect score of 0.00° is theoretically impossible — human tremor always produces some noise, and that noise is what the game measures.

5.3 Practice Mode

Practice mode allows unlimited attempts with no leaderboard submission. It is clearly visually separated from competitive mode. New users are directed to practice mode during onboarding. Practice attempts use the same angle as the day's competitive angle so players can calibrate before committing their one attempt.

6. Anti-Cheat System

The most important design constraint: the game must be impossible to win by placing the phone on a fixed surface. In the PWA version, tremor validation logic moves server-side (Supabase Edge Functions) to prevent client-side tampering.

Layer	Mechanism	Platform Note
1 — Shake gate	DeviceMotion API must register >0.8 m/s ² RMS for 2–3 seconds before angle reveals	Works on iOS Safari and Android Chrome
2 — Angle reveal timing	Angle shown for 1.5–2 seconds before targeting UI appears — no time to place phone	Extended from 1s in v1.0 for browser timing variance
3 — Micro-tremor validation	Server-side (Edge Function) checks: tremor SD $< 0.01^\circ$ for > 0.5 s = invalid flag	Moved from client (CoreMotion) to server for PWA security
4 — Single attempt lock	One attempt per 24-hour cycle per authenticated account	Unchanged from v1.0
5 — Orientation constraint	Portrait only, within 30° of vertical. Near-flat angles excluded from seed pool	Unchanged from v1.0

7. Daily Angle System

The daily angle is identical for every player worldwide, generated deterministically from the calendar date using a seeded pseudo-random number generator. The server validates the angle to prevent spoofing.

7.1 Generation Rules

- Range: 15° to 75° (excludes near-horizontal and near-vertical extremes)
- Axis: alternates between pitch and roll on a weekly schedule
- No repeat within a 30-day rolling window
- Seed formula: $\text{SHA-256}(\text{"steadyhand"} + \text{YYYY-MM-DD}) \bmod \text{valid_angle_range}$

7.2 Reveal Schedule

- New angle activates at 00:00 UTC daily
- Players receive a push notification at their local morning time (8:00 AM, configurable) — requires PWA install
- Angle is visible on the home screen before tapping Start

8. Leaderboard Architecture

8.1 Three-Tier Structure

Tier	Scope	Metric	Resets
Daily	Global — all players	Lowest MAD (ties share rank equally)	Midnight UTC
Weekly	Rolling 7-day window	Lowest average MAD across all 7 attempts	Every Monday 00:00 UTC
All-Time	Cumulative	Cumulative gold → silver → bronze	Never

8.2 Rank Display

All leaderboard positions are shown as percentile ranks (e.g. "top 8%") rather than absolute positions. This keeps the display motivating at any player count and eliminates the cold-start visibility problem on launch day.

8.3 Friend Leaderboards

- Up to 5 people: free from launch
- Up to 20 people: SteadyHand Pro subscription

Friend leaderboards are a primary daily retention driver. Moving basic friend groups into the free tier reflects the core loop's dependency on social competition.

8.4 End-of-Day Medal Ceremony

At midnight UTC, the daily leaderboard freezes. A push notification is sent to all players who submitted an attempt that day. Ranked players receive medal notification; others see their percentile finish. A shareable static card is generated for each medal winner — the primary viral sharing mechanism.

9. Profile & Progression

9.1 Profile Elements

- Display name + avatar (Apple or Google profile photo, optional)
- Medal cabinet: gold / silver / bronze counts
- Personal best score (all-time lowest MAD)
- Current streak: consecutive days with a valid attempt
- Attempts vs medals ratio
- Rank tier badge

9.2 Rank Tiers

Tier	Required Medals	Badge
Bronze	1–4 total medals	Bronze
Silver	5–14 total medals	Silver
Gold	15–29 total medals	Gold
Platinum	30–49 total medals	Platinum
Diamond	50+ total medals	Diamond

10. UX & Interface Design

The UI philosophy is tension through simplicity. The design language is dark-first, precision-instrument — near-black background, single ice-blue accent (#4FC3F7), monospace typography (JetBrains Mono) for all numeric readouts, and the dial as the centrepiece of the targeting and hold screens.

10.1 Screen Inventory

Screen	Key Elements
Home	Date, ??° hidden angle, personal best, leaderboard top 3 + your percentile, START button
Shake gate	Motion ring with fill progress, locked angle panel (hidden), move-phone instruction
Angle reveal	Large angle number (1.5–2s flash), pitch/roll axis label, countdown copy
Targeting	Live dial, current angle readout, $\pm 2^\circ$ tolerance arc highlighted in blue, lock indicator
Hold	Hold timer bar + countdown, live deviation readout, locked state indicator, subtle audio on drift
Score	Large score number, percentile rank, medal badges, deviation chart, share button
Leaderboard	Daily / weekly / all-time tabs, percentile ranks, top 100
Profile	Medal cabinet, stats, rank tier badge, streak

10.2 Audio Design (Replaces Haptics on iOS)

iOS Safari blocks the Web Vibration API. Audio cues serve as the primary feedback channel on iOS; Android retains real haptics as a supplementary layer.

Event	Audio Cue	Android Haptic
Shake gate active	Subtle low tone looping	Light continuous rumble
Angle reveal	Single mid tone	Medium impact
Lock acquired	Two-tone ascending chime	Double tap
Drift out of range	Soft descending pulse	Single soft pulse
Hold complete	Ascending resolution chord	Long strong impact
Medal awarded	Triple ascending tone	Triple impact pattern

11. Technical Architecture

11.1 Stack Overview

Layer	Technology	Rationale
Sensors	DeviceOrientation + DeviceMotion Web APIs	Browser equivalents of CoreMotion. ~60Hz on iOS Safari and Android Chrome.
Frontend	React + PWA (Service Worker)	Single codebase for iOS and Android. Offline support and Add-to-Home-Screen.
Auth	Sign in with Apple (OAuth) + Google OAuth	Apple retained for iOS users; Google added for Android users.
Backend	Supabase (PostgreSQL)	Real-time leaderboard queries. Unchanged from v1.0.
Anti-cheat	Supabase Edge Functions	Tremor validation moved server-side for PWA security.
Push	Web Push API	Requires PWA install. Replaces APNs from v1.0.
Audio	Web Audio API	Haptic replacement for iOS Safari.
Vibration	Web Vibration API	Android Chrome only. Supplementary to audio.
Analytics	PostHog (self-hosted)	Unchanged from v1.0.
Hosting	Vercel	Edge CDN, preview deploys, environment variables.

11.2 Sensor Specification

- API: DeviceOrientationEvent (tilt) + DeviceMotionEvent (acceleration)
- Update rate: ~60Hz (browser-capped; down from 100Hz CoreMotion in v1.0)
- Data used: event.beta (pitch) and event.gamma (roll), converted to degrees
- iOS permission: DeviceOrientationEvent.requestPermission() required on iOS 13+, triggered by user gesture during onboarding
- Tremor threshold: standard deviation of 0.1-second rolling window, sampled at 50Hz
- Surface detection: tremor SD < 0.01° for > 0.5s triggers invalid flag (validated server-side)

11.3 Backend Schema

- users — id, display_name, provider (apple|google), created_at, streak, total_attempts
- daily_angles — date (PK), angle_degrees, axis (pitch|roll)
- attempts — id, user_id, date, score_mad, submitted_at, valid, tremor_flag, device_model
- medals — id, user_id, date, medal_type (gold|silver|bronze)

- friend_groups — id, name, created_by, max_members (5 free, 20 pro)
- leaderboard_daily — materialised view, refreshed on insert to attempts

12. Notifications

Push notifications require the user to install SteadyHand as a PWA. Users on iOS who have not installed the PWA receive in-app banners only.

Trigger	Message	Timing
Daily angle drop	Today's angle is live. One shot. Good luck.	8:00 AM local (configurable)
EOD — medal winner	You placed today with 0.34°. Check your medal.	00:01 UTC
EOD — non-winner	Today's winner held 0.34°. Tomorrow's angle drops in Xh.	00:01 UTC
Streak reminder	You're on a 7-day streak. Don't break it.	6:00 PM if no attempt
Streak broken	Your streak ended. Reset and rebuild.	00:01 UTC if no attempt

13. Monetisation Strategy

The MVP launches free with no ads and no paywalls. Competitive integrity of the leaderboard depends on all players having identical experiences. Monetisation is introduced only after the core competitive loop is validated.

13.1 Phase 1 — Free (Launch)

- No ads, no IAP, no premium tier
- Focus: grow daily active users and leaderboard depth
- Basic friend groups (up to 5 people) free from launch

13.2 Phase 2 — Cosmetic IAP

- Custom dial skins (does not affect gameplay)
- Custom medal frame designs for share card
- Profile badge customisation
- All purely cosmetic — zero competitive advantage

13.3 Phase 3 — SteadyHand Pro Subscription

- Extended stats and historical angle archive
- Device comparison (how your phone model performs vs others)
- Private friend leaderboards for up to 20 people
- Early access to experimental game modes (dual-axis challenges)

14. Success Metrics

Metric	Target (90 days post-launch)	Measurement
Daily Active Users	5,000+	PostHog DAU
D7 Retention	> 40%	Cohort analysis
D30 Retention	> 20%	Cohort analysis
Avg session length	< 90 seconds	DeviceMotion event timestamps
Attempt rate	> 70% of DAU attempt each day	Supabase attempts table
Share card shares	> 15% of scorers share	UTM-tagged share sheet
Invalid attempt rate	< 3%	tremor_flag column
PWA install rate	> 25% of returning users	beforeinstallprompt event
App Store / Play rating	N/A — PWA	App store reviews (future)

15. Development Milestones

Phase	Milestone	Timeline
1	Core mechanic: DeviceOrientation API, angle targeting, timer + scoring	Week 1–2
2	iOS permission flow, PWA manifest + service worker, Add to Home Screen	Week 2–3
3	Anti-cheat: tremor detection (Edge Function), shake gate, one-attempt lock	Week 3–4
4	Daily angle seed, practice mode, local leaderboard, profile + medals	Week 4–6
5	Global leaderboard (Supabase), Apple + Google OAuth, Web Push API	Week 6–8
6	Audio design system, Android haptics, EOD medal ceremony, share card	Week 8–9
7	Polish: animations, friend groups, percentile ranks	Week 9–10
8	Beta testing, browser compatibility, performance audit, PWA submission	Week 10–12

16. Risks & Mitigations

Risk	Severity	Mitigation
iOS permission friction (motion requires user gesture)	High	Design onboarding prompt carefully. Show value proposition before asking permission. Fall back gracefully if denied.
Browser sensor rate (~60Hz vs 100Hz native)	Medium	Recalibrate tremor thresholds in beta. 50Hz sampling still sufficient for MAD scoring.
Surface-cheat via vibration motor trick	Low	Multi-layer anti-cheat with server-side validation. Flag outlier scores for manual review.
Leaderboard cold-start	High	Percentile ranks solve perception problem. Seed with 50–100 beta tester scores before public launch.
PWA push notifications require install step	Medium	In-app banners as fallback. Prompt install after first completed attempt.
Daily habit fails to form	High	Streak system + morning push (post-install). 30-second session feels ritualistic.
One-attempt rule causes frustration	Medium	Practice mode available before committing attempt. Clearly communicated in onboarding.

17. Out of Scope (v2.0)

- Native iOS app — PWA only for launch
- Native Android app — PWA only for launch
- Multiplayer / real-time head-to-head — asynchronous leaderboard only
- Custom challenge creation — daily angle is global and fixed
- Wearable integration (Apple Watch / Wear OS) — phone-only for MVP
- Dual-axis challenges (pitch AND roll simultaneously) — single axis per day
- Social graph / follow system — anonymous global leaderboard only
- In-app purchases — introduced post-validation only

18. Open Questions

18.1 Hold Duration

10 seconds is the current default. Should this vary by difficulty tier (easy 5s, standard 10s, hard 15s)?

Proposal: fixed at 10s for v2.0 to keep scoring comparable across the global leaderboard.

18.2 Tolerance Window

$\pm 2^\circ$ is current. Is this too easy for most players or too hard for casual users given browser sensor noise?

Proposal: run closed beta on both iOS and Android and measure mean scores; adjust before public launch, accounting for ~60Hz browser sampling.

18.3 Sensor Normalisation

Gyroscope variance differs between device models and OS versions. How do we ensure fair scoring across iPhone 12 through 16 Pro and a range of Android devices?

Proposal: collect device model in attempts table; analyse per-model distributions in beta; apply normalisation factor if variance exceeds a threshold.

18.4 Android App (Future)

Should a future v3.0 include a native Android app using SensorManager for higher sensor fidelity?

Proposal: defer — validate PWA audience size first. Revisit at 10k DAU.

End of Document

SteadyHand PRD v2.0 · June 2026 · Confidential